respectively as

a) $\left(3.25 \times 10^{-2} \text{ mol } L^{-1}s^{-1}\right)$ and $\left(1.3 \times 10^{-2} \text{ mol } L^{-1}s^{-1}\right)$

b) $\left(1.3 \times 10^{-2} \text{ mol } L^{-1}s^{-1}\right)$ and $\left(3.25 \times 10^{-2} \text{ mol } L^{-1}s^{-1}\right)$

c) $\left(1.3 \times 10^{-1} \text{ mol } L^{-1}s^{-1}\right)$ and $\left(3.25 \times 10^{-2} \text{ mol } L^{-1}s^{-1}\right)$

d) None of these

18. During the decomposition of $H_2O_2$ to give dioxygen, 48 g $O_2$ is formed per minute at certain point of time. The rate of formation of water at this point is

a) 0.75 mol $min^{-1}$

b) 1.5 mol $min^{-1}$

c) 2.25 mol $min^{-1}$

d) 3.0 mol $min^{-1}$

19. If the initial concentration of the reactant is doubled, the time for half reaction is also doubled. Then the order of the reaction is

a) Zero b) one

c) Fraction d) none

20. In a homogeneous reaction $A \longrightarrow B + C + D$, the initial pressure was $P_0$ and after time t it was P. expression for rate constant in terms of $P_0$, P and t will be

a) $k = \left(\frac{2.303}{t}\right)\log\left(\frac{2P_0}{3P_0 - P}\right)$

b) $k = \left(\frac{2.303}{t}\right)\log\left(\frac{2P_0}{P_0 - P}\right)$

c) $k = \left(\frac{2.303}{t}\right)\log\left(\frac{3P_0 - P}{2P_0}\right)$

d) $k = \left(\frac{2.303}{t}\right)\log\left(\frac{2P_0}{3P_0 - 2P}\right)$

21. If 75% of a first order reaction was completed in 60 minutes , 50% of the same reaction under the same conditions would be completed in

a) 20 minutes b) 30 minutes

c) 35 minutes d) 75 minutes

22. The half life period of a radioactive element is 140 days. After 560 days , 1 g of element will be reduced to

a) $\left(\frac{1}{2}\right)g$ b) $\left(\frac{1}{4}\right)g$

c) $\left(\frac{1}{8}\right)g$ d) $\left(\frac{1}{16}\right)g$

23. The correct difference between first and second order reactions is that (NEET)

a) A first order reaction can be catalysed; a second order reaction cannot be catalysed.

b) The half life of a first order reaction does not depend on [A0]; the half life of a second order reaction does depend on [A0].

c) The rate of a first order reaction does not depend on reactant concentrations; the rate of a second order reaction does depend on reactant concentrations.

d) The rate of a first order reaction does depend on reactant concentrations; the rate of a second order reaction does not depend on reactant concentrations.

24. After 2 hours, a radioactive substance becomes $\left(\frac{1}{16}\right)^{th}$ of original amount.



XII U7 kinetics - Jerald Folder.indd 229 3/29/2019 6:37:51 PM